KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 9 STANDARD

CHAPTER 10

YEAR 9 MATHEMATICS

# Surface area and volume

10 lessons · three pages each · Easy drill, then Medium and worded problems


10.1 Metric units
10.2 Precision and absolute error
10.3 Percentage error
10.4 Perimeter and area of composite shapes
10.5 Areas of quadrilaterals
10.6 Circumference and arc length
10.7 Areas of circles and sectors
10.8 Surface area of a prism
10.9 Surface area of a cylinder
10.10 Volumes of prisms and cylinders


NAME _____ CLASS _____ TEACHER _____

Artwork: a silver spiral on the western edge of Seahorse Valley, coloured in purple and silver, part of the Mandelbrot set.

## How to use this booklet

1. Each lesson is **three pages**. Read the **learning intention**, copy your teacher's notes into the **Notes** box, and tick each **success criterion** when you can do it.

2. Pages 1 and 2 are **Easy**: small steps, one round at a time. Page 3 is **Medium**, then three **worded problems**.

**WE DO** **Grey boxes.** Your teacher works each example on the board. **Copy every step** into the space.

**YOU DO** **Blue boxes.** Your turn. The **?** on each diagram shows what to find. Write the **formula**, **substitute** the numbers, then the **answer** with its unit.

3. Some Medium questions come from earlier lessons. The tag (like **10.2** or **Year 8**) says which one.

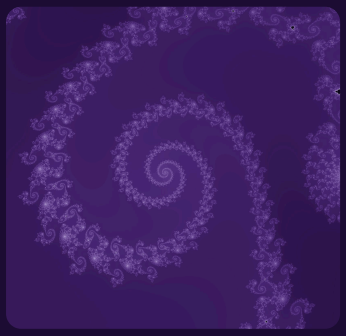
4. Stuck? Read the **Remember** line in the blue box, then look back at your notes and the example. Answers with π: **one decimal place** unless the question says otherwise.

## Contents and progress tracker


| Lesson | Topic | Page | Date | Done |
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| 10.2 | Precision and absolute error | 6 | | ☐ |
| 10.3 | Percentage error | 9 | | ☐ |
| 10.4 | Perimeter and area of composite shapes | 12 | | ☐ |
| 10.5 | Areas of quadrilaterals | 15 | | ☐ |
| 10.6 | Circumference and arc length | 18 | | ☐ |
| 10.7 | Areas of circles and sectors | 21 | | ☐ |
| 10.8 | Surface area of a prism | 24 | | ☐ |
| 10.9 | Surface area of a cylinder | 27 | | ☐ |
| 10.10 | Volumes of prisms and cylinders | 30 | | ☐ |


## In this chapter you will

- convert between metric units, including very large and very small units
- find the precision, absolute error and percentage error of a measurement
- find the perimeter and area of composite shapes and quadrilaterals
- find the circumference, arc length and area of circles, sectors and composite shapes
- find the surface area of prisms and cylinders
- find the volume and capacity of prisms and cylinders



### About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \to z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is a silver spiral on the western edge of Seahorse Valley, coloured in purple and silver. Maths can make beautiful things.

### Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

# 10.1 Metric units

EASY

**Learning intention** We are learning to convert between metric units, including time and digital storage.

**Success criteria**
- ☐ I can convert units of length, mass and capacity
- ☐ I can convert units of time and digital storage
- ☐ I can use the prefixes mega-, giga- and micro-

## NOTES — Key ideas for this lesson

TEACHER COPY

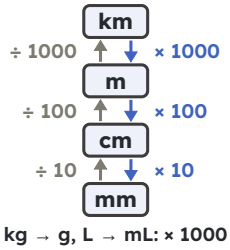
### KEY TERMS

**kilo- (k):** a thousand: 1 km = 1000 m
**centi- (c):** a hundredth: 1 cm = 0.01 m
**milli- (m):** a thousandth: 1 mm = 0.001 m
**micro- (μ):** a millionth: 1 μm = 0.000 001 m
**mega- (M):** a million: 1 ML = 1 000 000 L
**giga- (G):** a billion: 1 GB = 1 000 000 000 B
**capacity:** how much a container holds

### FORMULAS

**Bigger unit → smaller unit: ×**
**Smaller unit → bigger unit: ÷**
**1 kg = 1000 g · 1 t = 1000 kg**
**1 L = 1000 mL · 1 kL = 1000 L**
**1 h = 60 min · 1 min = 60 s**

**STEPS** 1 bigger or smaller unit? 2 write the rule 3 work it out 4 write the new unit

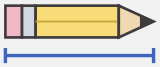



**kg → g, L → mL: × 1000**

## WE DO — With your teacher: copy what they write on the board.

**Example 1** Convert each one. To a smaller unit, multiply. To a bigger unit, divide.

**a** Length




**b** Length




**c** Mass




**d** Capacity



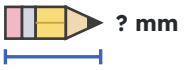

## YOU DO — On your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

**REMEMBER** To a smaller unit, multiply · To a bigger unit, divide · 1 km = 1000 m · 1 m = 100 cm · 1 cm = 10 mm

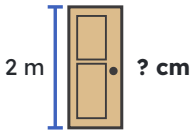
### ROUND 1 — Questions 1–4: convert each length to the smaller unit.

**1**



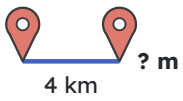

Rule: × 10
Working:
Answer:

**2**



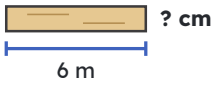

Rule: × 100
Working:
Answer:

**3**




Rule: × 1000
Working:
Answer:

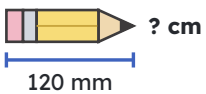
**4**




Rule: × 100
Working:
Answer:

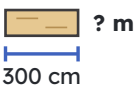
### ROUND 2 — Questions 5–8: convert each length to the bigger unit.

**5**



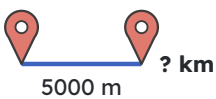

Rule: ÷ 10
Working:
Answer:

**6**



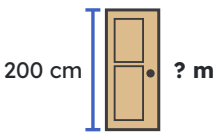

Rule: ÷ 100
Working:
Answer:

**7**




Rule: ÷ 1000
Working:
Answer:

**8**




Rule: ÷ 100
Working:
Answer:

# 10.1 Metric units

EASY

**YOU DO** On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

**REMEMBER** 1 kg = 1000 g · 1 L = 1000 mL · 1 min = 60 s · 1 h = 60 min · To a smaller unit, ×. To a bigger unit, ÷.

## ROUND 3 Questions 9–11: convert each mass.

**9**




Rule × 1000
Working
Answer

**10**




Rule ÷ 1000
Working
Answer

**11**




Rule × 1000
Working
Answer

## ROUND 4 Questions 12–14: convert each capacity.

**12**



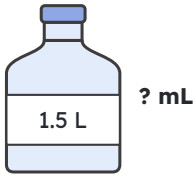

Rule × 1000
Working
Answer

**13**




Rule ÷ 1000
Working
Answer

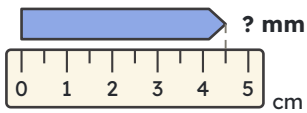
**14**




Rule × 1000
Working
Answer

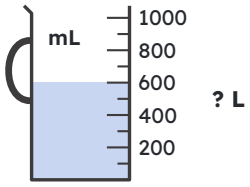
## ROUND 5 Questions 15–17: × or ÷? Write the rule, then convert. Read the scales carefully.

**15**




Rule
Working
Answer

**16**




Rule
Working
Answer

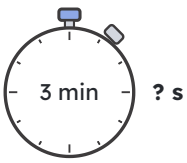
**17**




Rule
Working
Answer

## ROUND 6 Questions 18–20: convert each time. 1 min = 60 s and 1 h = 60 min.

**18**




Rule × 60
Working
Answer

**19**




Rule × 60
Working
Answer

**20**




Rule ÷ 60
Working
Answer

# 10.1 Metric units

MEDIUM

**WE DO** With your teacher: copy what they write on the board.

**Example 2** **Convert each one. Write the rule first.**

**a** Time



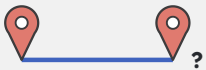

**? min**

**b** Storage




**? MB**

**c** Two steps




**? cm**

**d** Micro



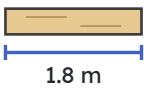
**? µm**




**YOU DO** On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

**REMEMBER** 1 GB = 1000 MB · 1 ML = 1 000 000 L · 1 g = 1000 mg · 1 mm = 1000 µm · m → mm: × 100, then × 10

**1** Length



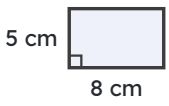

**? mm**

Rule _____

Working _____

Answer _____

**2** Area Year 8

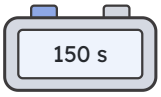



Formula _____

Substitute _____

Answer _____

**3** Time




**? min**

Rule _____

Working _____

Answer _____

**4** Round 4.0862 m to two decimal places. Year 8

Answer _____

**5** Storage



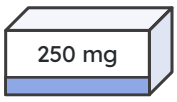

**? MB**

Rule _____

Working _____

Answer _____

**6** Mass



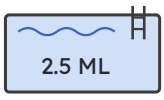

**? g**

Rule _____

Working _____

Answer _____

**7** Capacity



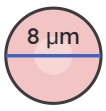

**? L**

Rule _____

Working _____

Answer _____

**8** Length




**? mm**

Rule _____

Working _____

Answer _____

**9** Find 15% of 60 kg. Year 8

Answer _____

**WORDED PROBLEMS** Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

**1** A phone has 64 GB of free space. Each video is 800 MB. How many videos fit on the phone?

Answer _____

**2** Mia runs laps of a 400 m track. How many laps does she run to cover 6 km?

Answer _____

**3** A movie runs for 135 minutes. It starts at 7:30 pm. At what time does it finish?

Answer _____

# 10.2 Precision and absolute error

EASY

**Learning intention** We are learning to find the precision, absolute error and limits of accuracy.

**Success criteria**
- ☐ I can find the precision of a scale or a measurement
- ☐ I can find the absolute error: precision ÷ 2
- ☐ I can write the limits of accuracy: m ± absolute error

## NOTES — Key ideas for this lesson

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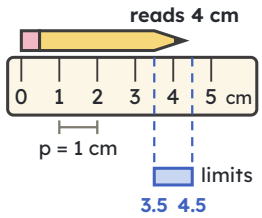
### KEY TERMS

**precision (p):** the size of one small space on a scale
**jump:** the difference between two numbers
**p of a measurement:** the place value of its last digit
**absolute error:** half the precision
**m:** the measurement
**true value:** the exact size, which we never know
**limits of accuracy:** the true value is between them

### FORMULAS

**p = jump ÷ spaces**
**p of 3.45 m = 0.01 m**
**Abs. error = p ÷ 2**
**Limits = m ± abs. error**

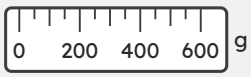
**STEPS** 1 find p 2 abs. error = p ÷ 2
3 lower = m − abs. error
4 upper = m + abs. error




## WE DO — With your teacher: copy what they write on the board.

**Example 1** Find the precision, the absolute error or the limits of accuracy.

**a** Precision




**b** Abs. error




**c** Abs. error




**d** Limits



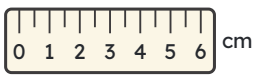

## YOU DO — On your own: work down the box, one round at a time. Show your working.

6 QUESTIONS

**REMEMBER** p = jump between two numbers ÷ number of spaces between them · Abs. error = p ÷ 2

### ROUND 1 — Questions 1–3: find the precision of each scale: the size of one small space.

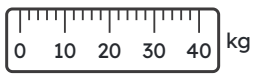
**1**




**precision = ?**

Formula: p = jump ÷ spaces
Substitute:
Answer:

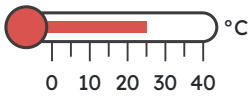
**2**




**precision = ?**

Formula: p = jump ÷ spaces
Substitute:
Answer:

**3**

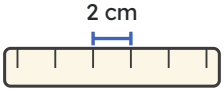



**precision = ?**

Formula: p = jump ÷ spaces
Substitute:
Answer:

### ROUND 2 — Questions 4–6: find the absolute error. The precision (one small space) is given.

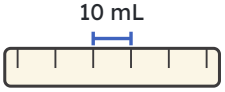
**4**




**abs. error = ?**

Formula: Abs. error = p ÷ 2
Substitute:
Answer:

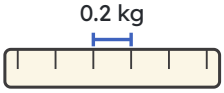
**5**




**abs. error = ?**

Formula: Abs. error = p ÷ 2
Substitute:
Answer:

**6**




**abs. error = ?**

Formula: Abs. error = p ÷ 2
Substitute:
Answer:

# 10.2 Precision and absolute error

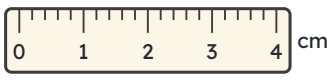
EASY

**YOU DO** On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

**REMEMBER** p = jump ÷ spaces · Abs. error = p ÷ 2 · p of 3.45 m = 0.01 m (its last digit) · Limits = m ± abs. error

## ROUND 3 Questions 7–9: find the absolute error of each scale.

**7**



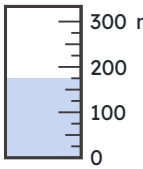

abs. error = ?

Formula: Abs. error = p ÷ 2

Substitute:

Answer:

**8**



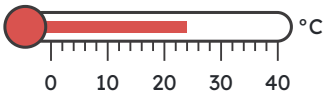

abs. error = ?

Formula: Abs. error = p ÷ 2

Substitute:

Answer:

**9**




abs. error = ?

Formula: Abs. error = p ÷ 2

Substitute:

Answer:

## ROUND 4 Questions 10–12: write the formula, then find the absolute error. The last digit gives p.

**10**



3.45 m

abs. error = ?

Formula:

Substitute:

Answer:

**11**




abs. error = ?

Formula:

Substitute:

Answer:

**12**




abs. error = ?

Formula:

Substitute:

Answer:

## ROUND 5 Questions 13–15: find the limits of accuracy of each measurement.

**13**



6 m

limits = ?

Formula: Limits = m ± abs. error

Substitute:

Answer:

**14**




limits = ?

Formula: Limits = m ± abs. error

Substitute:

Answer:

**15**




limits = ?

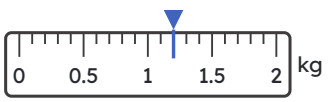
Formula: Limits = m ± abs. error

Substitute:

Answer:

## ROUND 6 Questions 16–18: read each scale, then find the limits of accuracy.

**16**



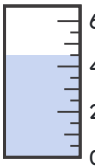

limits = ?

Formula: Limits = m ± abs. error

Substitute:

Answer:

**17**



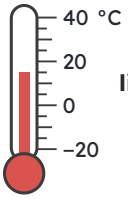

limits = ?

Formula: Limits = m ± abs. error

Substitute:

Answer:

**18**




limits = ?

Formula: Limits = m ± abs. error

Substitute:

Answer:

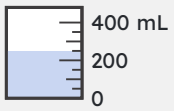
# 10.2 Precision and absolute error

MEDIUM

**WE DO** With your teacher: copy what they write on the board.

**Example 2** **Find the limits of accuracy of each measurement.**

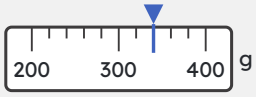
**a** Limits




**b** Limits




**c** Limits



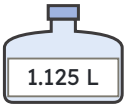

**d** Could it be 3.46 kg?




**YOU DO** On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

**REMEMBER** Lower limit = m − abs. error · Upper limit = m + abs. error · The true value is between the limits

**1** Abs. error




Formula _____

Substitute _____

Answer _____

**2** Convert 10.1




**= ? mm**

Rule _____

Working _____

Answer _____

**3** Limits

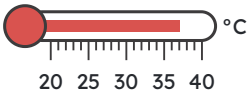



Formula _____

Substitute _____

Answer _____

**4** Limits

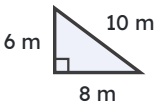



Formula _____

Substitute _____

Answer _____

**5** Area Year 8

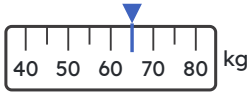



Formula _____

Substitute _____

Answer _____

**6** Limits




Formula _____

Substitute _____

Answer _____

**7** Convert 10.1




**= ? min**

Rule _____

Working _____

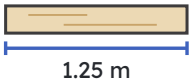
Answer _____

**8** These are the limits of accuracy of a length. Find the length and its absolute error.




Answer _____

**9** Could this plank really be 1.256 m long? Explain.




Answer _____

**WORDED PROBLEMS** Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

**1** A nurse's thermometer is marked every 0.2 °C. It shows 37.4 °C. A fever is 37.5 °C or above. Could the patient have a fever? Explain.

Answer _____

**2** Two planks are each 1.5 m long, to the nearest 0.1 m. They are laid end to end. Find the smallest and the largest possible total length.

Answer _____

**3** Sam ran 100 m in 13.4 s, timed to the nearest 0.1 s. Ali ran it in 13.36 s. Can we be sure that Ali was faster? Explain.

Answer _____

# 10.3 Percentage error

EASY

**Learning intention** We are learning to find percentage errors and use them to compare accuracy.

**Success criteria**
- ☐ I can find the absolute error: half the precision
- ☐ I can calculate percentage error = E ÷ M × 100%
- ☐ I can decide which measurement is more accurate

## NOTES — Key ideas for this lesson

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### KEY TERMS

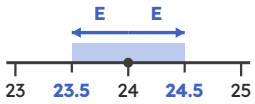
**precision (p):** the place value of the last digit, or one mark on a scale
**absolute error (E):** half the precision: the most the measurement can be out by
**measurement (M):** the value that was measured
**percentage error:** E as a percentage of M
**more accurate:** the smaller percentage error

### FORMULAS

**E = p ÷ 2**
**% error = E ÷ M × 100%**
**Limits: M − E to M + E**
**Smaller % error: more accurate**

**STEPS** 1 find p 2 E = p ÷ 2 3 E ÷ M × 100 4 round + %

**24 m, nearest m**




E = 1 ÷ 2 = 0.5 m
% error = 0.5 ÷ 24 × 100
**= 2.1%**

## WE DO — With your teacher: copy what they write on the board.

**Example 1** Find each absolute error or percentage error, to one decimal place where needed.

**a** Absolute error




**E = ?**

**b** % error



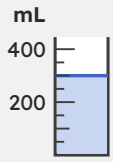

E = 0.5 m
**% error = ?**

**c** % error




**% error = ?**

**d** % error




**% error = ?**

## YOU DO — On your own: work down the box, one round at a time. Show your working.

6 QUESTIONS

**REMEMBER** p: the place value of the last digit (3.6 kg → 0.1 kg) · E = p ÷ 2 · % error = E ÷ M × 100

### ROUND 1 — Questions 1–3: find the absolute error E of each measurement.

**1**




**E = ?**
nearest m

Formula **E = p ÷ 2**
Substitute
Answer

**2**




**E = ?**

Formula **E = p ÷ 2**
Substitute
Answer

**3**

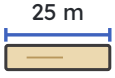



**E = ?**

Formula **E = p ÷ 2**
Substitute
Answer

### ROUND 2 — Questions 4–6: find the percentage error. The absolute error E is given.

**4**




E = 0.5 m
**% error = ?**

Formula **% error = E ÷ M × 100**
Substitute
Answer

**5**




E = 0.5 kg
**% error = ?**

Formula **% error = E ÷ M × 100**
Substitute
Answer

**6**




E = 0.05 L
**% error = ?**

Formula **% error = E ÷ M × 100**
Substitute
Answer

# 10.3 Percentage error

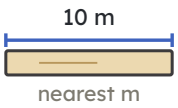
EASY

**YOU DO** On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

**REMEMBER** E = p ÷ 2 · % error = E ÷ M × 100 · On a scale, p = the gap between two marks · Round to one decimal place.

## ROUND 3 Questions 7–9: find the percentage error. Find E first.

**7**



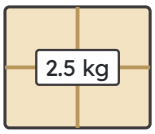

% error = ?

Formula % error = E ÷ M × 100

E = p ÷ 2 _____

Answer _____

**8**



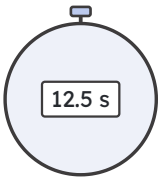

% error = ?

Formula % error = E ÷ M × 100

E = p ÷ 2 _____

Answer _____

**9**




% error = ?

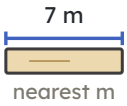
Formula % error = E ÷ M × 100

E = p ÷ 2 _____

Answer _____

## ROUND 4 Questions 10–12: write the formula, then find the percentage error.

**10**



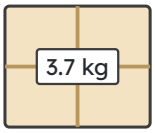

% error = ?

Formula _____

E = p ÷ 2 _____

Answer _____

**11**



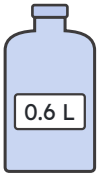

% error = ?

Formula _____

E = p ÷ 2 _____

Answer _____

**12**




% error = ?

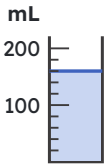
Formula _____

E = p ÷ 2 _____

Answer _____

## ROUND 5 Questions 13–15: read the scale, then find the percentage error. p is one mark.

**13**



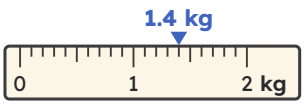

% error = ?

Formula % error = E ÷ M × 100

E = p ÷ 2 _____

Answer _____

**14**



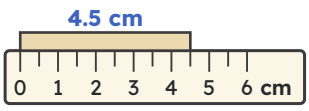

% error = ?

Formula % error = E ÷ M × 100

E = p ÷ 2 _____

Answer _____

**15**




% error = ?

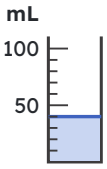
Formula % error = E ÷ M × 100

E = p ÷ 2 _____

Answer _____

## ROUND 6 Questions 16–18: write the formula, then find the percentage error.

**16**



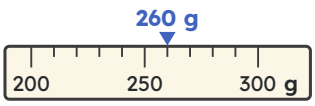

% error = ?

Formula _____

E = p ÷ 2 _____

Answer _____

**17**



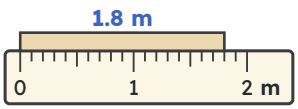

% error = ?

Formula _____

E = p ÷ 2 _____

Answer _____

**18**




% error = ?

Formula _____

E = p ÷ 2 _____

Answer _____

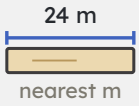
# 10.3 Percentage error

MEDIUM

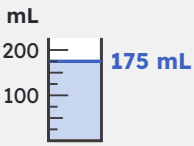
**WE DO** With your teacher: copy what they write on the board.

**Example 2** Give each answer correct to two decimal places.

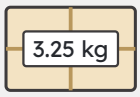
**a** % error



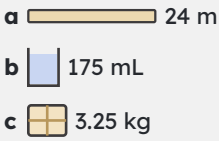

**b** % error




**c** % error




**d** Most accurate?




**YOU DO** On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

**REMEMBER** % error = E ÷ M × 100, to two decimal places · On a scale, p = one mark · Smaller % error: more accurate

**1** % error




Formula _____

E = p ÷ 2 _____

Answer _____

**2** Convert 10.1

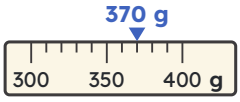



Rule _____

Working _____

Answer _____

**3** % error




Formula _____

E = p ÷ 2 _____

Answer _____

**4** % error

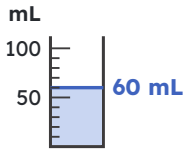



Formula _____

E = p ÷ 2 _____

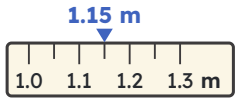
Answer _____

**5** Write the limits of accuracy. 10.2




Answer _____

**6** % error




Formula _____

E = p ÷ 2 _____

Answer _____

**7** More accurate? A 6 m B 12 m

% error A _____

% error B _____

Answer _____

**8** Abs. error 10.2




Formula _____

Substitute _____

Answer _____

**9** More accurate? A 0.75 kg B 2.5 kg

% error A _____

% error B _____

Answer _____

**WORDED PROBLEMS** Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

**1** Mia's pencil is 15 cm long and she is 168 cm tall, both to the nearest cm. Which measurement is more accurate? Give both percentage errors to two decimal places.

Answer _____

**2** A 100 m sprint is timed as 12.4 s. Find the percentage error of the time, correct to two decimal places.

Answer _____

**3** A jug marked every 50 mL shows 750 mL. Kai says the percentage error is 6.67%. What mistake did he make? Find the correct percentage error.

Answer _____

# 10.4 Perimeter and area of composite shapes

EASY

**Learning intention** We are learning to find the perimeter and area of composite shapes.

**Success criteria**
- ☐ I can find missing side lengths, then the perimeter
- ☐ I can find an area by splitting a shape into parts
- ☐ I can find a shaded area by subtracting

## NOTES — Key ideas for this lesson

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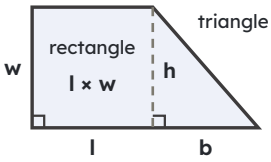
### KEY TERMS

**composite shape:** simple shapes joined together
**perimeter (P):** the distance around the outside
**area (A):** the space inside (cm², m²)
**missing side:** use the opposite sides: whole − part
**split:** cut the shape into parts with a dashed line
**height (h):** at right angles to the base (b)
**shaded area:** the whole shape − the white hole

### FORMULAS

**Rectangle: A = l × w**
**Triangle: A = ½ × b × h**
**Split: A = part 1 + part 2**
**Shaded: A = whole − hole**
**Perimeter: add every outside side**

**STEPS** 1 find missing sides
2 split, or subtract 3 area of each part
4 add or subtract + unit

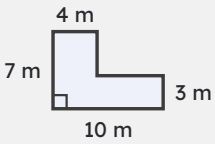



A = l × w + ½ × b × h

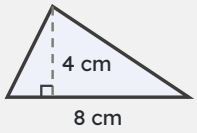
## WE DO — With your teacher: copy what they write on the board.

**Example 1** Find the perimeter or area. Find any missing sides first.

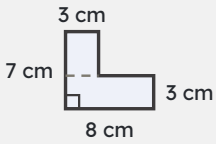
**a** Perimeter



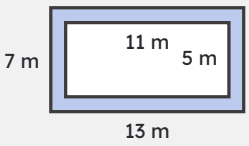

**b** Area




**c** Area




**d** Shaded area



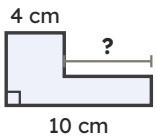

## YOU DO — On your own: work down the box, one round at a time. Show your working.

6 QUESTIONS

**REMEMBER** Missing side = whole side − the other part · Perimeter: add every side, the missing ones too.

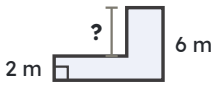
### ROUND 1 — Questions 1–3: find the missing side.

**1**



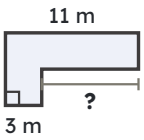

Formula: ? = whole − part
Substitute:
Answer:

**2**




Formula: ? = whole − part
Substitute:
Answer:

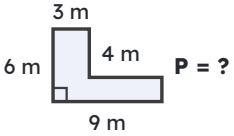
**3**




Formula: ? = whole − part
Substitute:
Answer:

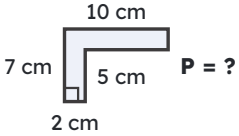
### ROUND 2 — Questions 4–6: find the perimeter. Find the two missing sides first.

**4**



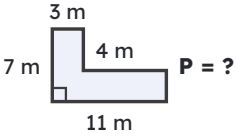

Formula: P = add all sides
Substitute:
Answer:

**5**




Formula: P = add all sides
Substitute:
Answer:

**6**



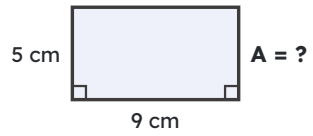

Formula: P = add all sides
Substitute:
Answer:

**YOU DO** On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

**REMEMBER** Rectangle A = l × w · Triangle A = ½ × b × h · Split: add the parts · Shaded: whole − hole

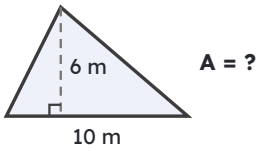
## ROUND 3 Questions 7–9: find the area of each rectangle or triangle.

**7**



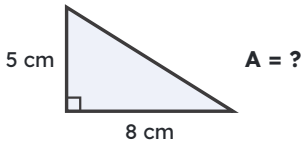

Formula A = l × w
Substitute
Answer

**8**




Formula A = ½ × b × h
Substitute
Answer

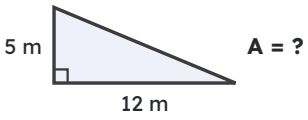
**9**




Formula A = ½ × b × h
Substitute
Answer

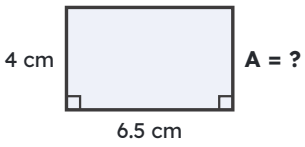
## ROUND 4 Questions 10–12: rectangle or triangle? Write the formula, then find the area.

**10**



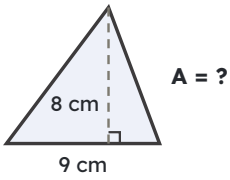

Formula
Substitute
Answer

**11**




Formula
Substitute
Answer

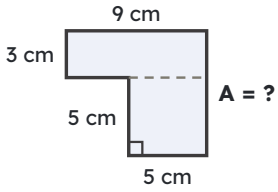
**12**




Formula
Substitute
Answer

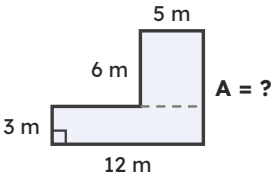
## ROUND 5 Questions 13–15: find the area. Split the shape, then add the parts.

**13**



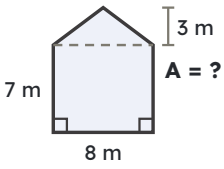

Formula A = l × w + l × w
Substitute
Answer

**14**




Formula A = l × w + l × w
Substitute
Answer

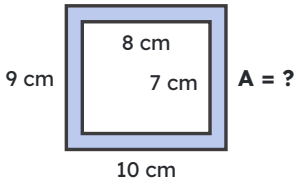
**15**




Formula A = l × w + ½ × b × h
Substitute
Answer

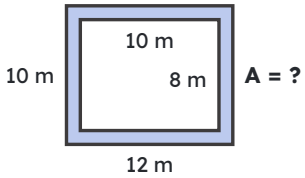
## ROUND 6 Questions 16–18: find each shaded area.

**16**



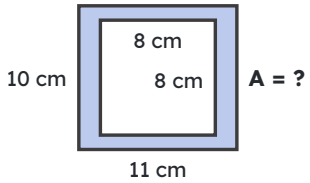

Formula A = whole − hole
Substitute
Answer

**17**




Formula A = whole − hole
Substitute
Answer

**18**




Formula A = whole − hole
Substitute
Answer

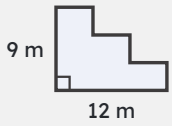
# 10.4 Perimeter and area of composite shapes

MEDIUM

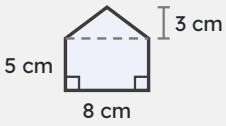
**WE DO** With your teacher: copy what they write on the board.

**Example 2** **Find the perimeter or area. Split the shape, or subtract.**

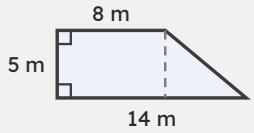
**a** Perimeter



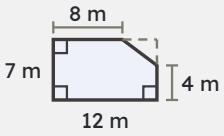

**b** Area




**c** Area



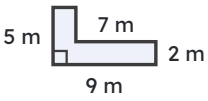

**d** Area




**YOU DO** On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

**REMEMBER** Find missing sides first · Split into rectangles and triangles, or subtract the part cut out

**1** Perimeter




Formula _____

Substitute _____

Answer _____

**2** In cm [10.1]

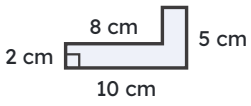



Rule _____

Working _____

Answer _____

**3** Area

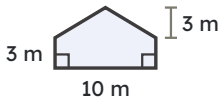



Formula _____

Substitute _____

Answer _____

**4** Area




Formula _____

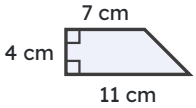
Substitute _____

Answer _____

**5** Write the limits of accuracy of 6.4 m. [10.2]

Answer _____

**6** Area

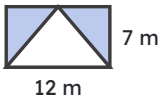



Formula _____

Substitute _____

Answer _____

**7** Shaded area

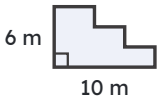



Formula _____

Substitute _____

Answer _____

**8** Perimeter




Formula _____

Substitute _____

Answer _____

**9** Find the percentage error of 40 cm, measured to the nearest cm. [10.3]

Formula _____

Substitute _____

Answer _____

**WORDED PROBLEMS** Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

**1** An L-shaped room is made of a 6 m by 4 m rectangle and a 3 m by 2 m rectangle. Carpet costs \$45 per m². Find the cost to carpet the room.

Answer _____

**2** An L-shaped garden bed has outside edges of 8 m and 6 m. Each arm of the L is 2 m wide. Edging costs \$15 per metre. Find the cost to edge the whole bed.

Answer _____

**3** The end wall of a shed is a rectangle 4 m wide and 2.5 m high, with a triangle 1.5 m high on top. It has a door 0.9 m wide and 2 m high. Find the area of the wall to paint.

Answer _____

# 10.5 Areas of quadrilaterals

EASY

**Learning intention** We are learning to find the area of a parallelogram, a trapezium, a rhombus and a kite.

**Success criteria**
- ☐ I can use A = bh for a parallelogram
- ☐ I can use A = ½(a + b)h for a trapezium
- ☐ I can use A = ½xy for a rhombus or a kite

## NOTES — Key ideas for this lesson

TEACHER COPY

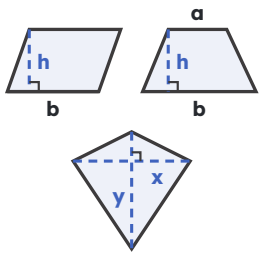
### KEY TERMS

**parallelogram:** two pairs of parallel sides
**trapezium:** one pair of parallel sides, a and b
**rhombus:** a parallelogram with 4 equal sides
**kite:** two pairs of equal sides, next to each other
**base (b):** the side the height is measured from
**height (h):** perpendicular (at right angles) to b
**diagonals (x, y):** join opposite corners

### FORMULAS

**Parallelogram: A = bh**
**Trapezium: A = ½(a + b)h**
**Rhombus or kite: A = ½xy**
**Missing length: substitute, then solve**

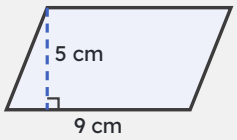
**STEPS** 1 name the shape
2 write the formula 3 substitute
4 work out + unit (cm²)



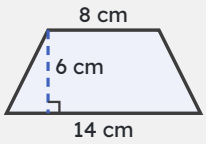

## WE DO — With your teacher: copy what they write on the board.

**Example 1** Find the area of each shape. Write the formula first.

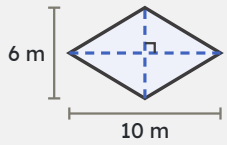
**a** Parallelogram



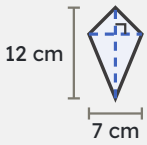

**b** Trapezium




**c** Rhombus




**d** Kite



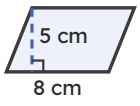

## YOU DO — On your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

**REMEMBER** h is the perpendicular height (the dashed line), at right angles to the base · a and b are the parallel sides.

### ROUND 1 — Questions 1–4: find the area of each parallelogram.

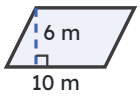
**1**




A = ?

Formula A = bh
Substitute
Answer

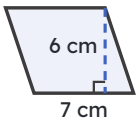
**2**




A = ?

Formula A = bh
Substitute
Answer

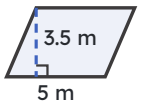
**3**




A = ?

Formula A = bh
Substitute
Answer

**4**

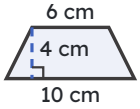



A = ?

Formula A = bh
Substitute
Answer

### ROUND 2 — Questions 5–8: find the area of each trapezium.

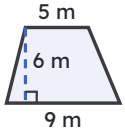
**5**




A = ?

Formula A = ½(a + b)h
Substitute
Answer

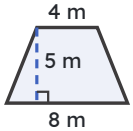
**6**




A = ?

Formula A = ½(a + b)h
Substitute
Answer

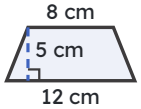
**7**




A = ?

Formula A = ½(a + b)h
Substitute
Answer

**8**




A = ?

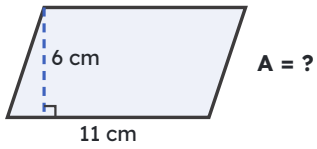
Formula A = ½(a + b)h
Substitute
Answer

**YOU DO** On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

**REMEMBER** Parallelogram A = bh · Trapezium A = ½(a + b)h · Rhombus or kite A = ½xy (x and y are the diagonals)

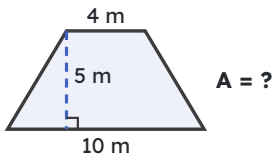
## ROUND 3 Questions 9–11: parallelogram or trapezium? Write the formula, then find the area.

**9**



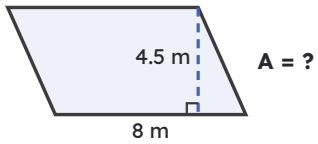

Formula _____
Substitute _____
Answer _____

**10**




Formula _____
Substitute _____
Answer _____

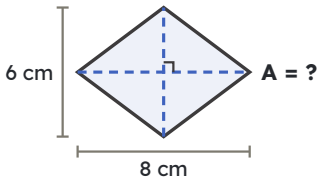
**11**




Formula _____
Substitute _____
Answer _____

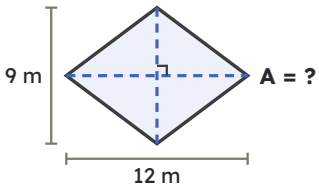
## ROUND 4 Questions 12–14: find the area of each rhombus.

**12**



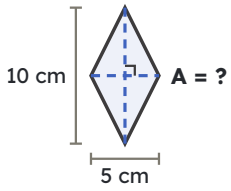

Formula A = ½xy
Substitute _____
Answer _____

**13**




Formula A = ½xy
Substitute _____
Answer _____

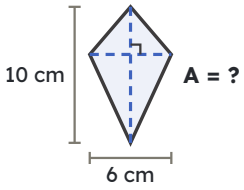
**14**




Formula A = ½xy
Substitute _____
Answer _____

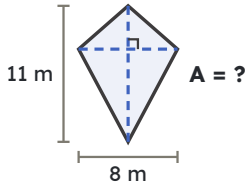
## ROUND 5 Questions 15–17: find the area of each kite.

**15**



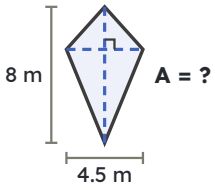

Formula A = ½xy
Substitute _____
Answer _____

**16**




Formula A = ½xy
Substitute _____
Answer _____

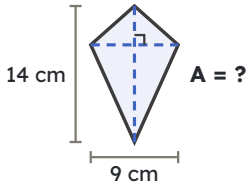
**17**




Formula A = ½xy
Substitute _____
Answer _____

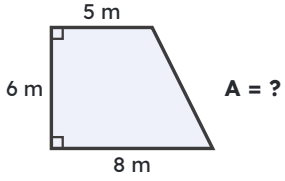
## ROUND 6 Questions 18–20: which shape is it? Write the formula, then find the area.

**18**



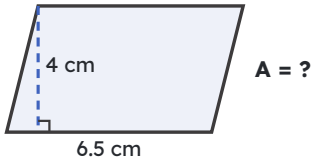

Formula _____
Substitute _____
Answer _____

**19**




Formula _____
Substitute _____
Answer _____

**20**

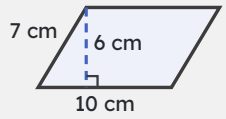



Formula _____
Substitute _____
Answer _____

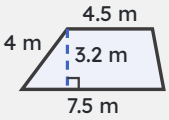
**WE DO** With your teacher: copy what they write on the board.

**Example 2** **Choose the right formula. Some lengths are not needed.**

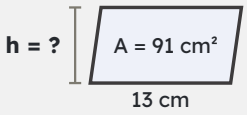
**a** Area



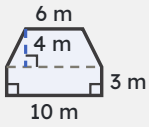

**b** Area




**c** Height



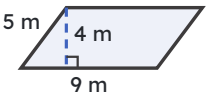

**d** Area




**YOU DO** On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

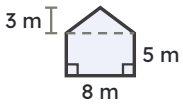
**REMEMBER** The slanted side is not the height · Missing h: substitute, then solve · Composite: add the parts

**1** Area



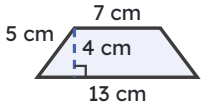

Formula _____
Substitute _____
Answer _____

**2** Area 10.4



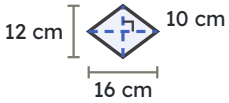

Formula _____
Substitute _____
Answer _____

**3** Area




Formula _____
Substitute _____
Answer _____

**4** Area

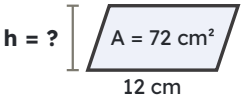



Formula _____
Substitute _____
Answer _____

**5** Find the percentage error of 25 kg, measured to the nearest kg. 10.3

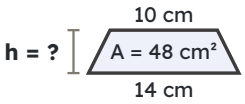
Formula _____
Substitute _____
Answer _____

**6** Height




Formula _____
Substitute _____
Answer _____

**7** Height

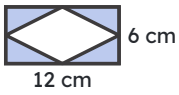



Formula _____
Substitute _____
Answer _____

**8** Write the limits of accuracy of 8.5 m. 10.2

Answer _____

**9** Shaded area




Formula _____
Substitute _____
Answer _____

**WORDED PROBLEMS** Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

**1** A courtyard is a parallelogram with base 8.5 m and perpendicular height 6 m. Pavers cost $42 per m². Find the cost of paving the courtyard.

Answer _____

**2** A kite is made from two rods, 80 cm and 50 cm long, that cross at right angles. Find the area of material needed to cover it.

Answer _____

**3** A block of land is a trapezium with parallel sides of 30 m and 45 m. Its area is 750 m². Find the distance between the parallel sides.

Answer _____

# 10.6 Circumference and arc length

EASY

**Learning intention** We are learning to find circumferences, arc lengths and perimeters of curved shapes.

**Success criteria**
- ☐ I can use C = πd or C = 2πr to find a circumference
- ☐ I can find an arc length: θ ÷ 360 × 2πr
- ☐ I can find the perimeter of sectors and composite shapes

## NOTES — Key ideas for this lesson

TEACHER COPY

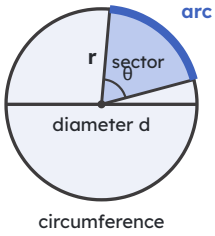
**KEY TERMS**

**circumference (C):** the distance around a circle
**radius (r):** from the centre to the edge
**diameter (d):** across through the centre; d = 2r
**arc:** part of the circumference
**sector:** a slice of a circle between two radii
**θ:** the angle at the centre of a sector
**perimeter:** the distance around a shape

**FORMULAS**

**C = πd = 2πr**
**Arc length = θ ÷ 360 × 2πr**
**Semicircle: P = ½πd + d**
**Quadrant: P = ¼ × 2πr + 2r**
**Sector: P = arc + 2r**

**STEPS** 1 r = d ÷ 2 2 write the formula 3 substitute (π key) 4 round to 1 d.p. + unit



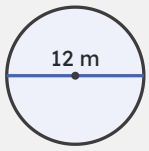

## WE DO — With your teacher: copy what they write on the board.

**Example 1** Find each circumference or perimeter, correct to one decimal place.

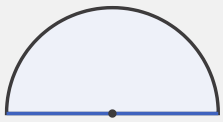
**a** Circumference



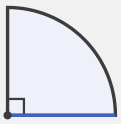

**b** Circumference




**c** Perimeter




**d** Perimeter



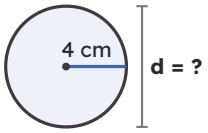

## YOU DO — On your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

**REMEMBER** d = 2r · r = d ÷ 2 · C = πd · Use the π key, then round to one decimal place.

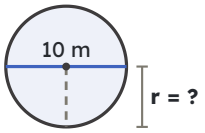
### ROUND 1 — Questions 1–4: find the diameter or the radius.

**1**



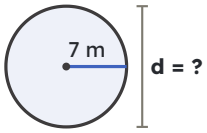

Formula: d = 2r
Substitute:
Answer:

**2**



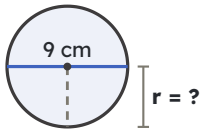

Formula: r = d ÷ 2
Substitute:
Answer:

**3**




Formula: d = 2r
Substitute:
Answer:

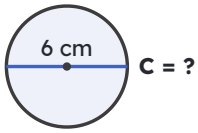
**4**




Formula: r = d ÷ 2
Substitute:
Answer:

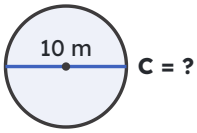
### ROUND 2 — Questions 5–8: find the circumference. The diameter is given.

**5**



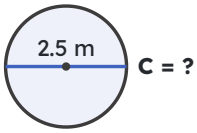

Formula: C = πd
Substitute:
Answer:

**6**



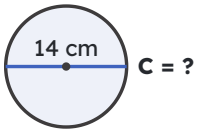

Formula: C = πd
Substitute:
Answer:

**7**




Formula: C = πd
Substitute:
Answer:

**8**



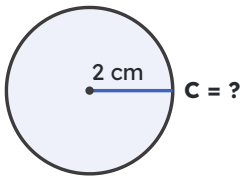

Formula: C = πd
Substitute:
Answer:

**YOU DO** On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

**REMEMBER** C = 2πr = πd · Semicircle P = ½πd + d · Quadrant P = ¼ × 2πr + 2r · Find d first when r is given.

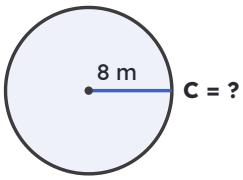
## ROUND 3 Questions 9–11: find the circumference. The radius is given.

**9**



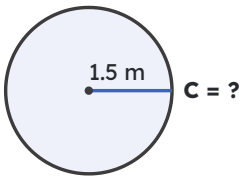

Formula C = 2πr
Substitute
Answer

**10**




Formula C = 2πr
Substitute
Answer

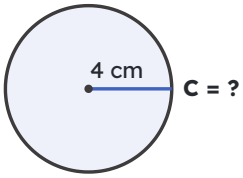
**11**




Formula C = 2πr
Substitute
Answer

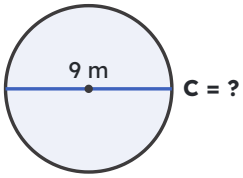
## ROUND 4 Questions 12–14: is it r or d? Write the right formula, then find the circumference.

**12**



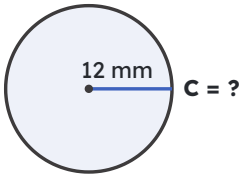

Formula
Substitute
Answer

**13**




Formula
Substitute
Answer

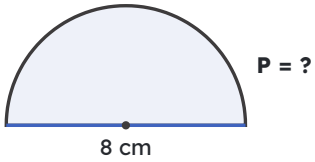
**14**




Formula
Substitute
Answer

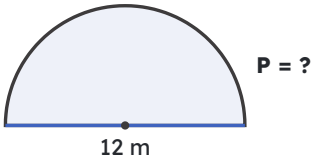
## ROUND 5 Questions 15–17: find the perimeter of each semicircle.

**15**



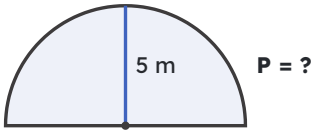

Formula P = ½πd + d
Substitute
Answer

**16**




Formula P = ½πd + d
Substitute
Answer

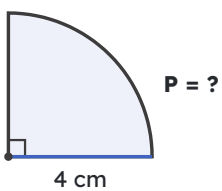
**17**




Formula P = ½πd + d
Substitute
Answer

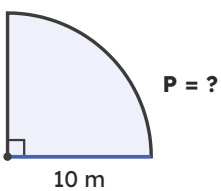
## ROUND 6 Questions 18–20: find the perimeter of each quadrant.

**18**



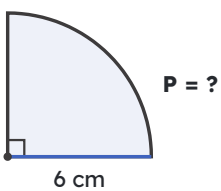

Formula P = ¼ × 2πr + 2r
Substitute
Answer

**19**




Formula P = ¼ × 2πr + 2r
Substitute
Answer

**20**




Formula P = ¼ × 2πr + 2r
Substitute
Answer

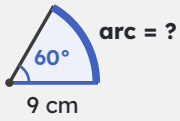
# 10.6 Circumference and arc length

MEDIUM

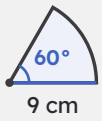
**WE DO** With your teacher: copy what they write on the board.

**Example 2** **Give each answer correct to one decimal place.**

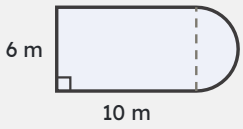
**a** Arc length



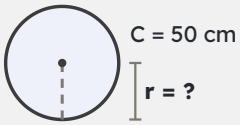

**b** Perimeter




**c** Perimeter




**d** Radius




**YOU DO** On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

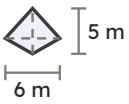
**REMEMBER** Arc = θ ÷ 360 × 2πr (θ = angle at the centre) · Sector P = arc + 2r · Composite: add the outside edges only

**1** Arc length



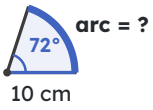

Formula _____
Substitute _____
Answer _____

**2** Area 10.5




Formula _____
Substitute _____
Answer _____

**3** Arc length



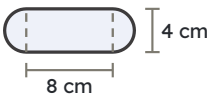

Formula _____
Substitute _____
Answer _____

**4** Perimeter




Formula _____
Substitute _____
Answer _____

**5** Perimeter




Formula _____
Substitute _____
Answer _____

**6** Perimeter

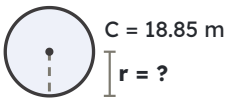



Formula _____
Substitute _____
Answer _____

**7** Write the limits of accuracy of 4.5 m. 10.2

Answer _____

**8** Radius




Formula _____
Substitute _____
Answer _____

**9** Find the percentage error of 20 cm, measured to the nearest cm. 10.3

Formula _____
Substitute _____
Answer _____

**WORDED PROBLEMS** Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

**1** A round garden bed has a diameter of 3.2 m. Edging costs $12 per metre. Find the cost to edge the bed, to the nearest dollar.

Answer _____

**2** A bike wheel has a radius of 33 cm. How many full turns does it make to travel 1 km?

Answer _____

**3** A running track is a rectangle 84 m long with a semicircle on each end. Each semicircle has a diameter of 64 m. Find one lap, to the nearest metre.

Answer _____

# 10.7 Areas of circles and sectors

EASY

**Learning intention** We are learning to find the areas of circles, semicircles, quadrants, sectors and composite shapes.

**Success criteria**
- ☐ I can use $A = \pi r^2$ to find the area of a circle
- ☐ I can find the area of a sector: $\theta \div 360 \times \pi r^2$
- ☐ I can find composite and shaded areas

## NOTES — Key ideas for this lesson

TEACHER COPY

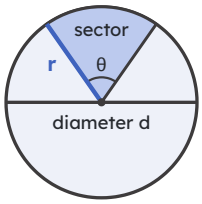
### KEY TERMS

**area (A):** the space inside a shape, in square units
**radius (r):** from the centre to the edge
**diameter (d):** across through the centre; $r = d \div 2$
**semicircle:** half a circle
**quadrant:** a quarter of a circle
**sector:** a slice between two radii, angle $\theta$
**shaded area:** the whole shape – the part cut out

### FORMULAS

**Circle:** $A = \pi r^2$
**Semicircle:** $A = \frac{1}{2}\pi r^2$
**Quadrant:** $A = \frac{1}{4}\pi r^2$
**Sector:** $A = \theta \div 360 \times \pi r^2$
**Ring:** $A = \pi R^2 - \pi r^2$ **(R outer, r inner)**

**STEPS** 1 $r = d \div 2$ 2 write the formula 3 substitute ($\pi$ key) 4 round to 1 d.p. + unit²



circle: $A = \pi r^2$

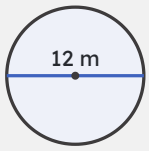
## WE DO — With your teacher: copy what they write on the board.

**Example 1** Find the area of each shape, correct to one decimal place.

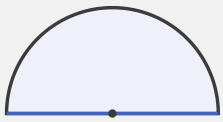
**a** Area



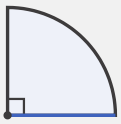

**b** Area




**c** Area




**d** Area




## YOU DO — On your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

**REMEMBER** $A = \pi r^2$ · $r = d \div 2$ · Use the $\pi$ key, then round to one decimal place · Area units: cm², m².

### ROUND 1 — Questions 1–4: find the area of each circle. The radius is given.

**1**



A = ?

Formula $A = \pi r^2$
Substitute _____
Answer _____

**2**



A = ?

Formula $A = \pi r^2$
Substitute _____
Answer _____

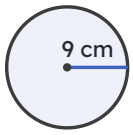
**3**



A = ?

Formula $A = \pi r^2$
Substitute _____
Answer _____

**4**

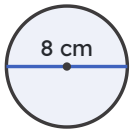


A = ?

Formula $A = \pi r^2$
Substitute _____
Answer _____

### ROUND 2 — Questions 5–8: find the area. The diameter is given: halve it first.

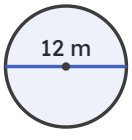
**5**



A = ?

Formula $A = \pi r^2$
Substitute _____
Answer _____

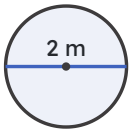
**6**



A = ?

Formula $A = \pi r^2$
Substitute _____
Answer _____

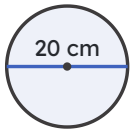
**7**



A = ?

Formula $A = \pi r^2$
Substitute _____
Answer _____

**8**



A = ?

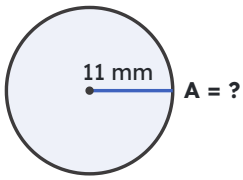
Formula $A = \pi r^2$
Substitute _____
Answer _____

**YOU DO** On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

**REMEMBER** $A = \pi r^2$ · Semicircle $A = \frac{1}{2}\pi r^2$ · Quadrant $A = \frac{1}{4}\pi r^2$ · Sector $A = \theta \div 360 \times \pi r^2$ · Halve d first.

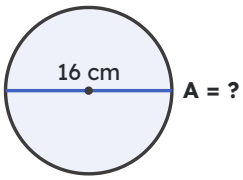
## ROUND 3 Questions 9–11: is it r or d? Write the formula, then find the area.

**9**



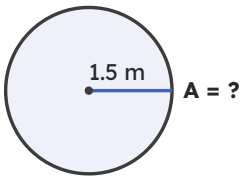

Formula _____
Substitute _____
Answer _____

**10**




Formula _____
Substitute _____
Answer _____

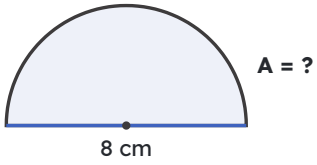
**11**




Formula _____
Substitute _____
Answer _____

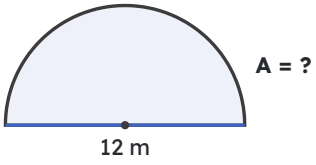
## ROUND 4 Questions 12–14: find the area of each semicircle.

**12**



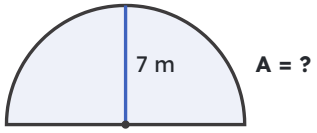

Formula $A = \frac{1}{2}\pi r^2$
Substitute _____
Answer _____

**13**




Formula $A = \frac{1}{2}\pi r^2$
Substitute _____
Answer _____

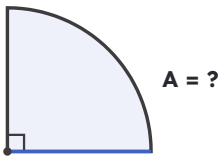
**14**




Formula $A = \frac{1}{2}\pi r^2$
Substitute _____
Answer _____

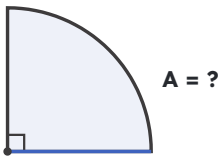
## ROUND 5 Questions 15–17: find the area of each quadrant.

**15**



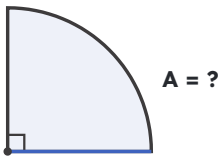

Formula $A = \frac{1}{4}\pi r^2$
Substitute _____
Answer _____

**16**




Formula $A = \frac{1}{4}\pi r^2$
Substitute _____
Answer _____

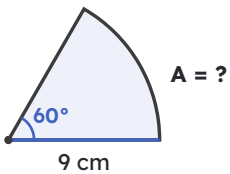
**17**




Formula $A = \frac{1}{4}\pi r^2$
Substitute _____
Answer _____

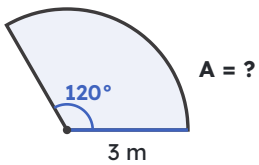
## ROUND 6 Questions 18–20: find the area of each sector.

**18**



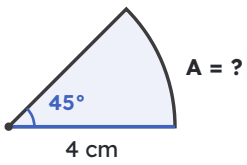

Formula $A = \theta \div 360 \times \pi r^2$
Substitute _____
Answer _____

**19**




Formula $A = \theta \div 360 \times \pi r^2$
Substitute _____
Answer _____

**20**




Formula $A = \theta \div 360 \times \pi r^2$
Substitute _____
Answer _____

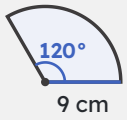
# 10.7 Areas of circles and sectors

MEDIUM

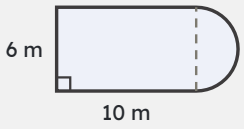
**WE DO** With your teacher: copy what they write on the board.

**Example 2** **Give each answer correct to one decimal place.**

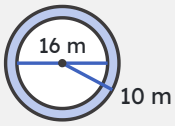
**a** Area



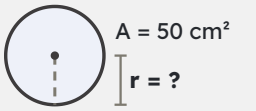

**b** Area




**c** Shaded area




**d** Radius




**YOU DO** On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

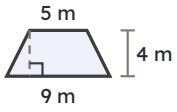
**REMEMBER** Sector A = θ ÷ 360 × πr² · Composite: add the parts · Shaded: whole − cut-out · Radius: r² = A ÷ π, then √

**1** Area



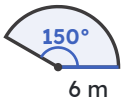

Formula _____
Substitute _____
Answer _____

**2** Area 10.5



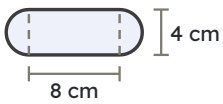

Formula _____
Substitute _____
Answer _____

**3** Area




Formula _____
Substitute _____
Answer _____

**4** Area



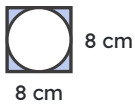

Formula _____
Substitute _____
Answer _____

**5** Perimeter 10.6



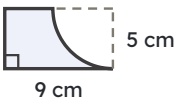

Formula _____
Substitute _____
Answer _____

**6** Shaded area



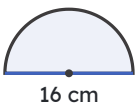

Formula _____
Substitute _____
Answer _____

**7** Area



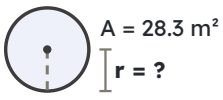

Formula _____
Substitute _____
Answer _____

**8** Perimeter 10.6




Formula _____
Substitute _____
Answer _____

**9** Radius




Formula _____
Substitute _____
Answer _____

**WORDED PROBLEMS** Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

**1** A garden sprinkler sprays water 6 m and turns through an angle of 120°. Find the area of lawn it waters.

Answer _____

**2** A round pond has a radius of 3 m. A path 1 m wide goes all the way around it. Find the area of the path.

Answer _____

**3** A pizza has a diameter of 30 cm. It is cut into 8 equal slices. Find the area of one slice.

Answer _____

# 10.8 Surface area of a prism

EASY

**Learning intention** We are learning to find the surface area of a prism by adding the areas of its faces.

**Success criteria**
- ☐ I can find the area of each face of a prism
- ☐ I can use SA = 6s² and SA = 2(lw + lh + wh)
- ☐ I can find the surface area of triangular prisms and open boxes

## NOTES — Key ideas for this lesson

TEACHER COPY

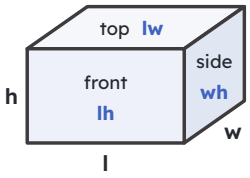
### KEY TERMS

**prism:** the same cross-section all the way along
**face:** a flat surface of a solid
**surface area (SA):** the total area of all the faces
**cube:** 6 equal square faces
**rectangular prism:** 3 pairs of equal rectangles
**triangular prism:** 2 triangles and 3 rectangles
**no lid:** an open box: leave out the top

### FORMULAS

**Cube: SA = 6s²**
**Rectangular prism: SA = 2(lw + lh + wh)**
**Triangular prism: SA = bh + perimeter × l**
**No lid: SA = lw + 2lh + 2wh**
**Any prism: 2 × end + perimeter × length**

**STEPS** 1 name the faces 2 area of each face 3 add them up 4 unit² (cm²)

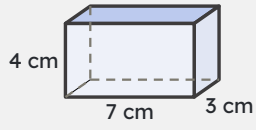



SA = 2(lw + lh + wh)

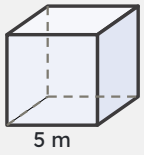
## WE DO — With your teacher: copy what they write on the board.

**Example 1** Find each area. Surface area (SA) is the total area of all the faces.

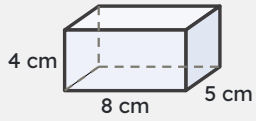
**a** Shaded face



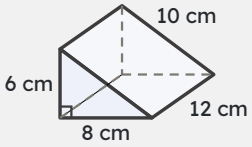

**b** Surface area




**c** Surface area




**d** Surface area



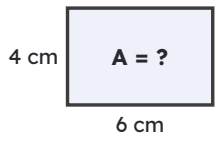

## YOU DO — On your own: work down the box, one round at a time. Show your working.

7 QUESTIONS

**REMEMBER** Each face is a rectangle: A = l × w · Front: l × h · Top: l × w · Side: w × h · Areas are in square units (cm²).

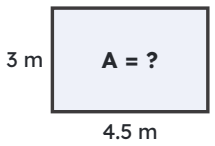
### ROUND 1 — Questions 1–4: find the area of each face.

**1**



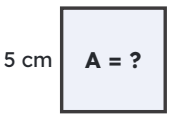

Formula A = l × w
Substitute
Answer

**2**



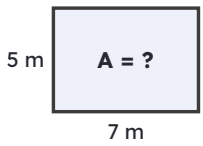

Formula A = l × w
Substitute
Answer

**3**




Formula A = l × w
Substitute
Answer

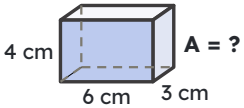
**4**




Formula A = l × w
Substitute
Answer

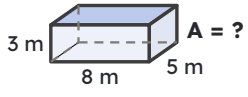
### ROUND 2 — Questions 5–7: find the area of the shaded face of each rectangular prism.

**5**



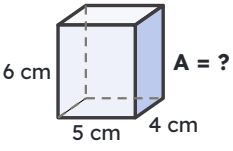

Formula A = l × h
Substitute
Answer

**6**




Formula A = l × w
Substitute
Answer

**7**




Formula A = w × h
Substitute
Answer

# 10.8 Surface area of a prism

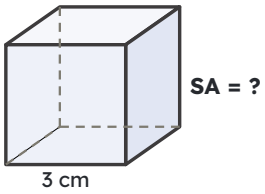
EASY

**YOU DO** On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

**REMEMBER** Cube: SA = $6s^2$ · Rectangular prism: SA = 2(lw + lh + wh) · Triangular prism: 2 triangles + 3 rectangles

## ROUND 3 Questions 8–10: find the surface area of each cube.

**8**

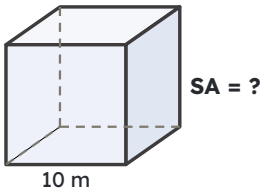



Formula SA = $6s^2$

Substitute

Answer

**9**

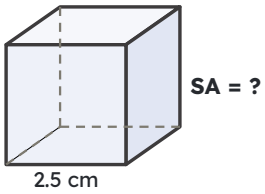



Formula SA = $6s^2$

Substitute

Answer

**10**



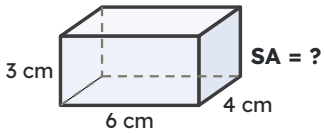

Formula SA = $6s^2$

Substitute

Answer

## ROUND 4 Questions 11–13: find the surface area of each rectangular prism.

**11**

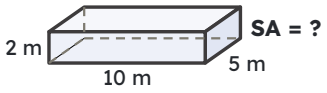



Formula SA = 2(lw + lh + wh)

Substitute

Answer

**12**

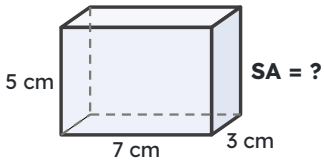



Formula SA = 2(lw + lh + wh)

Substitute

Answer

**13**



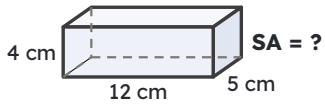

Formula SA = 2(lw + lh + wh)

Substitute

Answer

## ROUND 5 Questions 14–16: cube or rectangular prism? Write the formula, then find the surface area.

**14**

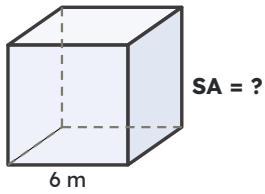



Formula

Substitute

Answer

**15**

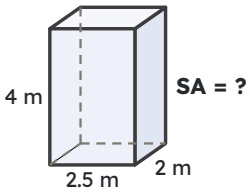



Formula

Substitute

Answer

**16**



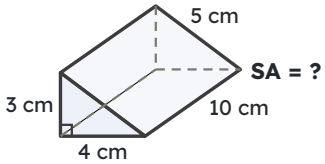

Formula

Substitute

Answer

## ROUND 6 Questions 17–19: find the surface area of each right-angled triangular prism.

**17**

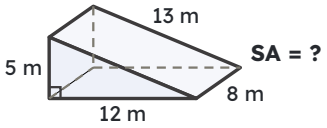



Formula SA = bh + (b + h + s)l

Substitute

Answer

**18**

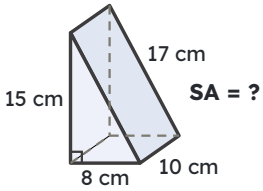



Formula SA = bh + (b + h + s)l

Substitute

Answer

**19**




Formula SA = bh + (b + h + s)l

Substitute

Answer

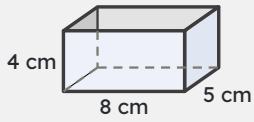
# 10.8 Surface area of a prism

MEDIUM

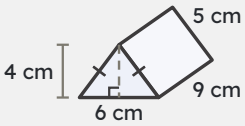
**WE DO** With your teacher: copy what they write on the board.

**Example 2** **Find each surface area. List the faces first.**

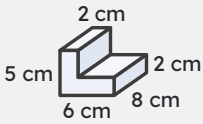
**a** SA (no lid)



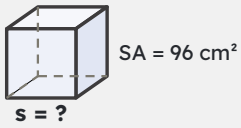

**b** Surface area




**c** Surface area



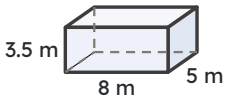

**d** Side length




**YOU DO** On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

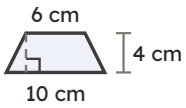
**REMEMBER** No lid: SA = lw + 2lh + 2wh · Any prism: 2 × end + perimeter × length · Cube: $s^2$ = SA ÷ 6 · π answers to 1 d.p.

**1** Surface area



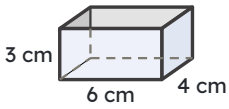

Formula _____
Substitute _____
Answer _____

**2** Area 10.5



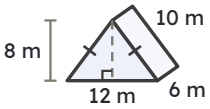

Formula _____
Substitute _____
Answer _____

**3** SA (no lid)




Formula _____
Substitute _____
Answer _____

**4** Surface area



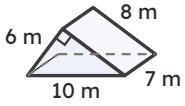

Formula _____
Substitute _____
Answer _____

**5** Area 10.7



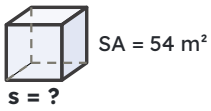

Formula _____
Substitute _____
Answer _____

**6** Surface area



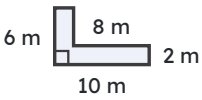

Formula _____
Substitute _____
Answer _____

**7** Side length



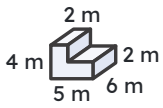

Formula _____
Substitute _____
Answer _____

**8** Area 10.4




Formula _____
Substitute _____
Answer _____

**9** Surface area




Formula _____
Substitute _____
Answer _____

**WORDED PROBLEMS** Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

**1** A gift box is 30 cm long, 20 cm wide and 10 cm high. Find the area of paper needed to cover it exactly.

Answer _____

**2** A room is 5 m long, 4 m wide and 2.5 m high. 1 L of paint covers 10 m². How many litres of paint are needed for the four walls and the ceiling?

Answer _____

**3** A tent is a triangular prism 4 m long. Each end is a triangle: base 3 m, height 2 m, sloping sides 2.5 m. Find the area of canvas, including the floor.

Answer _____

# 10.9 Surface area of a cylinder

EASY

**Learning intention** We are learning to find the surface area of a cylinder: two circles and a rolled-up rectangle.

**Success criteria**
- ☐ I can find the area of a circular end: $A = \pi r^2$
- ☐ I can find the curved surface area: $A = 2\pi rh$
- ☐ I can find the surface area of closed and open cylinders

## NOTES — Key ideas for this lesson

TEACHER COPY

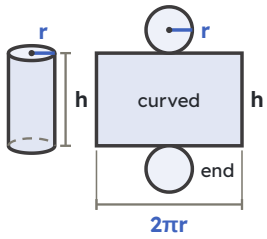
**KEY TERMS**

**cylinder:** a solid with two equal circular ends
**radius (r):** centre of an end to its edge; r = d ÷ 2
**height (h):** the distance between the two ends
**curved surface:** the side; flat, a 2πr by h rectangle
**net:** the flat shape that folds up into the solid
**surface area (SA):** the total area of all the surfaces
**no lid:** an open cylinder: one end only

**FORMULAS**

**End: $A = \pi r^2$**
**Curved surface: $A = 2\pi rh$**
**Closed: $SA = 2\pi r^2 + 2\pi rh$**
**No lid: $SA = \pi r^2 + 2\pi rh$**
**Half cylinder: $SA = \pi r^2 + \pi rh + 2rh$**

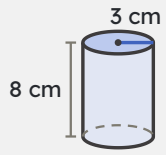
**STEPS** 1 r = d ÷ 2 2 list the surfaces 3 substitute (π key) 4 round to 1 d.p. + unit²



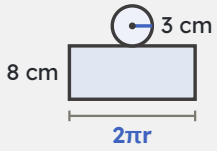

## WE DO — With your teacher: copy what they write on the board.

**Example 1** Find each area, correct to one decimal place.

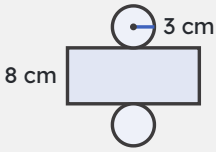
**a** End area



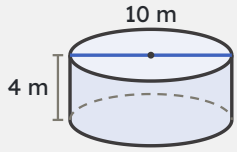

**b** Curved surface




**c** Surface area




**d** Surface area



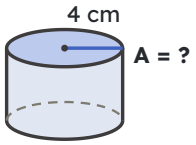

## YOU DO — On your own: work down the box, one round at a time. Show your working.

7 QUESTIONS

**REMEMBER** End: $A = \pi r^2$ · Curved surface: $A = 2\pi rh$ (h is the height) · Use the π key · Round to one decimal place.

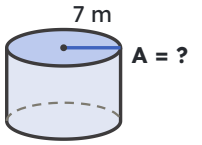
### ROUND 1 — Questions 1–4: find the area of the shaded end of each cylinder.

**1**



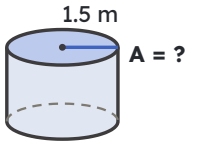

Formula A = πr²
Substitute
Answer

**2**



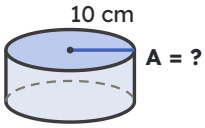

Formula A = πr²
Substitute
Answer

**3**




Formula A = πr²
Substitute
Answer

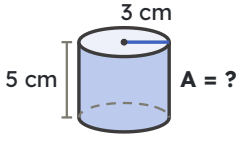
**4**




Formula A = πr²
Substitute
Answer

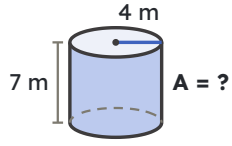
### ROUND 2 — Questions 5–7: find the curved surface area (shaded).

**5**



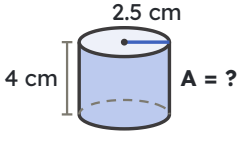

Formula A = 2πrh
Substitute
Answer

**6**




Formula A = 2πrh
Substitute
Answer

**7**




Formula A = 2πrh
Substitute
Answer

# 10.9 Surface area of a cylinder

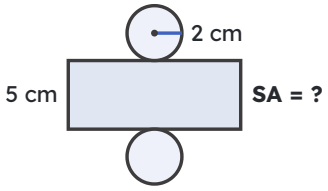
EASY

**YOU DO** On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

**REMEMBER** Closed cylinder: SA = 2πr² + 2πrh (two ends + the curved surface) · r = d ÷ 2 · Round to one decimal place.

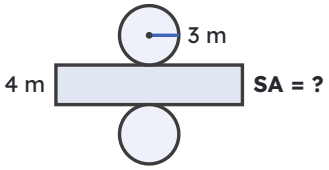
## ROUND 3 Questions 8–10: find the total area of each net. It is the surface area of the cylinder.

**8**



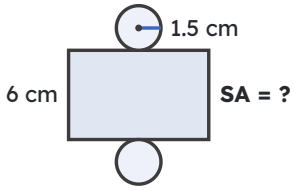

Formula SA = 2πr² + 2πrh
Substitute
Answer

**9**




Formula SA = 2πr² + 2πrh
Substitute
Answer

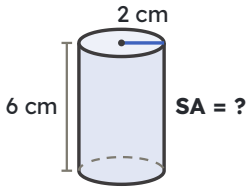
**10**




Formula SA = 2πr² + 2πrh
Substitute
Answer

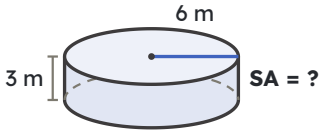
## ROUND 4 Questions 11–13: find the surface area of each closed cylinder. The radius is given.

**11**



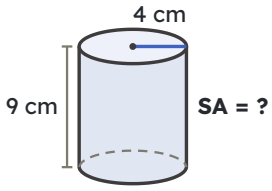

Formula SA = 2πr² + 2πrh
Substitute
Answer

**12**




Formula SA = 2πr² + 2πrh
Substitute
Answer

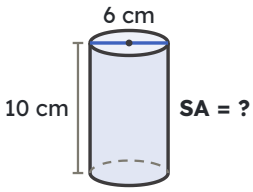
**13**




Formula SA = 2πr² + 2πrh
Substitute
Answer

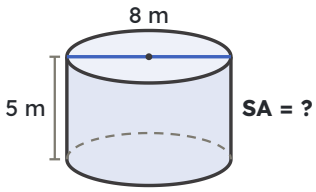
## ROUND 5 Questions 14–16: find the surface area. The diameter is given: halve it first.

**14**



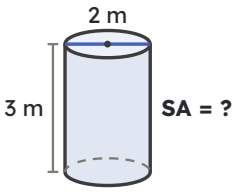

Formula SA = 2πr² + 2πrh
Substitute
Answer

**15**




Formula SA = 2πr² + 2πrh
Substitute
Answer

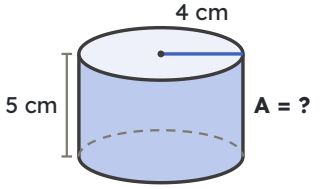
**16**




Formula SA = 2πr² + 2πrh
Substitute
Answer

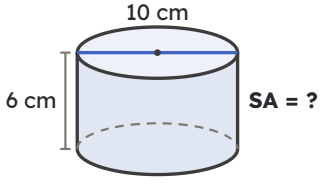
## ROUND 6 Questions 17–19: which surface? Write the formula, then find the area.

**17**



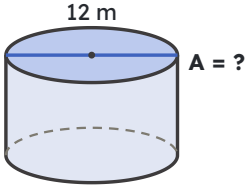

Formula
Substitute
Answer

**18**




Formula
Substitute
Answer

**19**




Formula
Substitute
Answer

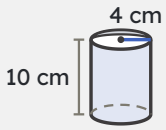
# 10.9 Surface area of a cylinder

MEDIUM

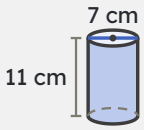
**WE DO** With your teacher: copy what they write on the board.

**Example 2** **Find each area, correct to one decimal place. List the surfaces first.**

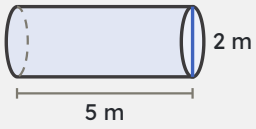
**a** SA (no lid)



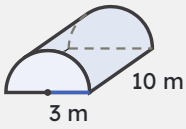

**b** Label area




**c** Surface area



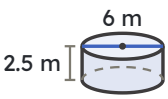

**d** Surface area




**YOU DO** On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

**REMEMBER** No lid: SA = $\pi r^2 + 2\pi rh$ · Label: $2\pi rh$ only · Half cylinder: SA = $\pi r^2 + \pi rh + 2rh$ · Lying down: h is the length

**1** Surface area




Formula _____
Substitute _____
Answer _____

**2** Perimeter 10.6



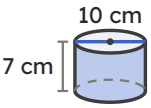

Formula _____
Substitute _____
Answer _____

**3** SA (no lid)



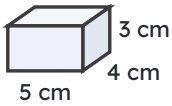

Formula _____
Substitute _____
Answer _____

**4** Label area



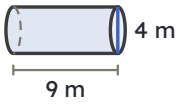

Formula _____
Substitute _____
Answer _____

**5** SA 10.8



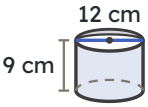

Formula _____
Substitute _____
Answer _____

**6** Surface area




Formula _____
Substitute _____
Answer _____

**7** SA (no lid)



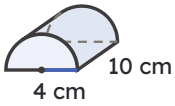

Formula _____
Substitute _____
Answer _____

**8** Area 10.7




Formula _____
Substitute _____
Answer _____

**9** Surface area




Formula _____
Substitute _____
Answer _____

**WORDED PROBLEMS** Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

**1** A closed cylindrical tank has a diameter of 2 m and a height of 1.5 m. Paint costs \$14 per m². Find the cost to paint the whole tank, to the nearest dollar.

Answer _____

**2** A paint roller is a cylinder 25 cm long with a diameter of 6 cm. Find the area it paints in 10 full turns.

Answer _____

**3** A bucket with no lid is a cylinder with a diameter of 30 cm and a height of 25 cm. Find the area of plastic needed to make it.

Answer _____

# 10.10 Volumes of prisms and cylinders

EASY

**Learning intention** We are learning to find the volume of a prism or a cylinder ($V = Ah$), and its capacity.

**Success criteria**
- ☐ I can use $V = Ah$ to find the volume of any prism
- ☐ I can use $V = \pi r^2 h$ to find the volume of a cylinder
- ☐ I can change between $cm^3$, mL, L and $m^3$

## NOTES — Key ideas for this lesson

TEACHER COPY

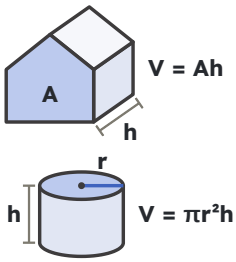
### KEY TERMS

**volume (V):** the space inside a solid, in cubic units
**prism:** the same cross-section all along
**cross-section (A):** the end of the prism (shaded)
**height of the prism (h):** the distance between the two ends
**cylinder:** a solid whose cross-section is a circle
**capacity:** how much a container holds (mL, L)

### FORMULAS

**$V = Ah$**
**Cuboid: $V = lwh$**
**Triangular prism: $V = \frac{1}{2}bh \times l$** (b, h: the triangle; l: the length)
**Cylinder: $V = \pi r^2 h$**
**$1\ cm^3 = 1\ mL \cdot 1\ m^3 = 1000\ L$**

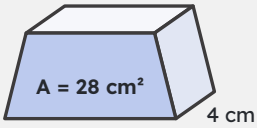
**STEPS** 1 find A, the end
2 write the formula 3 substitute (π key)
4 answer in $cm^3$ or $m^3$



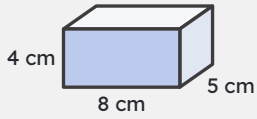

## WE DO — With your teacher: copy what they write on the board.

**Example 1** Find the volume of each solid. Give the cylinder correct to one decimal place.

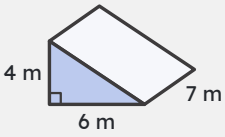
**a** Prism



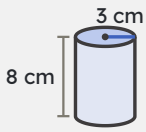

**b** Cuboid




**c** Triangular prism




**d** Cylinder



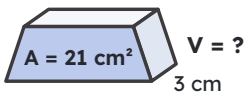

## YOU DO — On your own: work down the box, one round at a time. Show your working.

6 QUESTIONS

**REMEMBER** $V = Ah$: A is the area of the shaded end, h the distance between the ends · Cuboid: $V = lwh$ · Units: $cm^3$, $m^3$

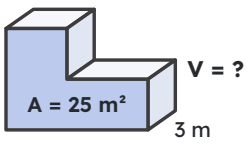
### ROUND 1 — Questions 1–3: find the volume. The area of the shaded end is given.

**1**



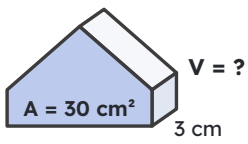

Formula V = Ah
Substitute
Answer

**2**




Formula V = Ah
Substitute
Answer

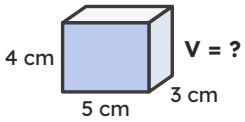
**3**




Formula V = Ah
Substitute
Answer

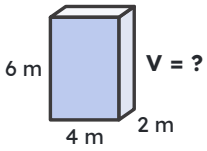
### ROUND 2 — Questions 4–6: find the volume of each cuboid.

**4**



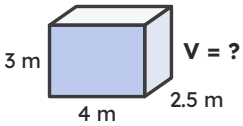

Formula V = lwh
Substitute
Answer

**5**




Formula V = lwh
Substitute
Answer

**6**




Formula V = lwh
Substitute
Answer

# 10.10 Volumes of prisms and cylinders

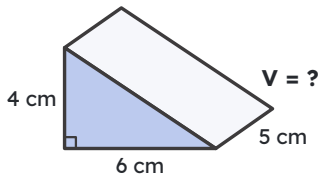
EASY

**YOU DO** On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

**REMEMBER** Triangular prism: V = ½bh × l (b, h: the triangle; l: the length) · Cylinder: V = πr²h · r = d ÷ 2 · Use the π key, then round to one decimal place.

## ROUND 3 Questions 7–9: find the volume of each triangular prism.

**7**

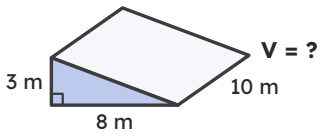



Formula V = ½bh × l

Substitute

Answer

**8**

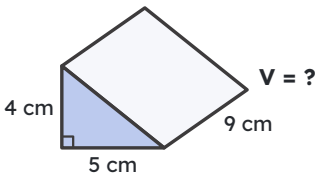



Formula V = ½bh × l

Substitute

Answer

**9**



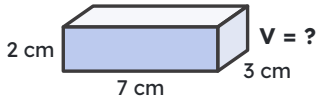

Formula V = ½bh × l

Substitute

Answer

## ROUND 4 Questions 10–12: which solid is it? Write its formula, then find the volume.

**10**

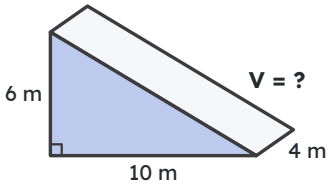



Formula

Substitute

Answer

**11**

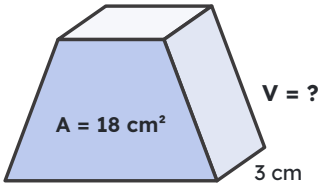



Formula

Substitute

Answer

**12**



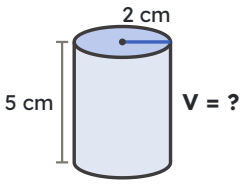

Formula

Substitute

Answer

## ROUND 5 Questions 13–15: find the volume of each cylinder. The radius is given.

**13**

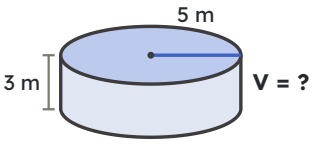



Formula V = πr²h

Substitute

Answer

**14**

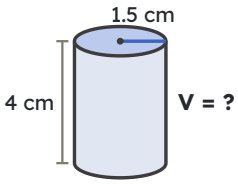



Formula V = πr²h

Substitute

Answer

**15**



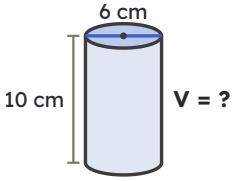

Formula V = πr²h

Substitute

Answer

## ROUND 6 Questions 16–18: find the volume. The diameter is given: halve it first.

**16**

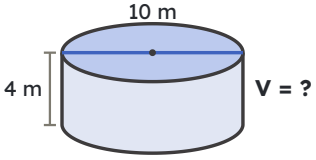



Formula V = πr²h

Substitute

Answer

**17**

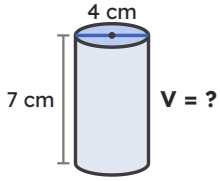



Formula V = πr²h

Substitute

Answer

**18**




Formula V = πr²h

Substitute

Answer

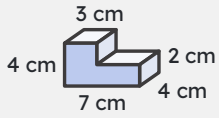
# 10.10 Volumes of prisms and cylinders

MEDIUM

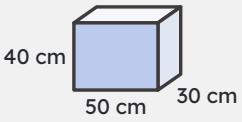
**WE DO** With your teacher: copy what they write on the board.

**Example 2** Round π answers to one decimal place. 1 cm³ = 1 mL, 1000 cm³ = 1 L, 1 m³ = 1000 L.

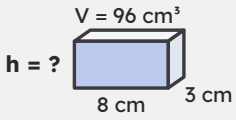
**a** Volume



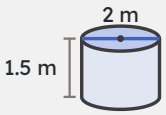

**b** Capacity in L




**c** Height



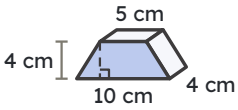

**d** Capacity in L




**YOU DO** On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

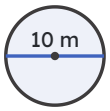
**REMEMBER** Find A, the end, first · L-shape: two rectangles · Trapezium: A = ½(a + b)h · Height: h = V ÷ (l × w)

**1** Volume



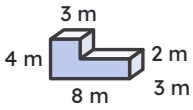

Formula _____
Substitute _____
Answer _____

**2** Area 10.7



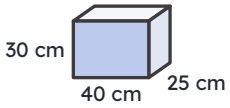

Formula _____
Substitute _____
Answer _____

**3** Volume



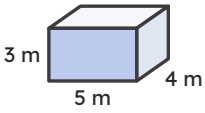

Formula _____
Substitute _____
Answer _____

**4** Capacity in L



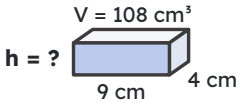

Formula _____
Substitute _____
Answer _____

**5** Surface area 10.8



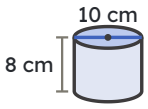

Formula _____
Substitute _____
Answer _____

**6** Height



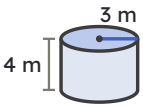

Formula _____
Substitute _____
Answer _____

**7** Capacity in mL



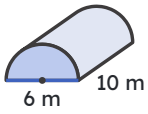

Formula _____
Substitute _____
Answer _____

**8** Surface area 10.9




Formula _____
Substitute _____
Answer _____

**9** Volume




Formula _____
Substitute _____
Answer _____

**WORDED PROBLEMS** Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

**1** A tent is a triangular prism. Its triangular end has a base of 2.4 m and a height of 1.5 m, and the tent is 3 m long. Find the volume of air inside the tent.

Answer _____

**2** A cylindrical rainwater tank has a diameter of 3 m and a height of 2 m. How many litres does it hold, to the nearest litre?

Answer _____

**3** A pool is 25 m long and 10 m wide. It holds 375 000 L of water. How deep is the water?

Answer _____